Aim:- Shell programming: Shell script exercises based on following

(i) Interactive shell scripts.
(ii) Positional parameters
(iii) Arithmetic
(iv) if then if, if-then-else if, nested if-else.
(v) logical operators
(vi) else + if equals elif, case structure.
(vii) While, Until, for loops, use of break
(viii) Metacharacters
(ix) System administration: disk management & daily administration.

(a) Write a shell scripting program to demonstrate on Interactive shell - script

```
#!/bin/bash
# Linux shell scripting (interactive shell script)

echo -n "Input First value"
read input1
echo -n "Input second value"
read input2

echo "Your First value:" $input1
echo "Your Second value:" $input2
```

Output

```
Input First value 34
Input second value 56

Your First value : 34
Your Second value : 56
```

output

sh exp3. sh hi hello how r u

Positional parameters are hi hello how r u

Output

Input First value : 12
Input Second value : 2
Addition of two Numbers : 14
Substraction of numbers : 10
Multiplication of number; 24
Division of two numbers : 6

(b) Write a shell scripting program to demonstrate positional parameters:

```
#!/bin/bash
#Linux shell scripting (Positional shell Script)

arg =0
temp_total =0
number_of_args = $#

#To see the sufficient commend line arguments
if [ $# -it 2]
then echo "OOPS! I need at least 2 commend line
argumenh."
else echo "Positional parameters are" $1 $2 $3
$4 $5
```

(c) Write a shell scripting program to demonstrate Arithmetic operation.

```
#!/bin/bash
#Linux Shell Scripting (Arithmetic Shell scripting)
echo -n " Input First Value:"
read n1
echo -n "Input Second value:"
read n2
a ='expr $n1 + $n2'
echo "Additional of two numbers :" $a
b = 'expr $n1 - $n2'
echo " Subtraction of numbers:" $b
```

output

Enter number :
456
Number is greater then 100..

output :

Input value to check: 35
number is between 1 & 100..

~~c = ' $~~

c = ' expr $n1 * $n2'
echo " Multiplication of numbers : " $c
d = ' expr $n1 / $n2'
echo " Division of two numbers : " $d

(d) Write a shell scripting program to demonstrate if then-fi, if-then-else, if ~~to~~ nested if-else.

(i)
```
#!/bin/bash
#Linux Shell Scripting (if-then-fi shell scripting)
echo "Enter number:"
read no
if [ $no -gt 100]
then echo "Number is greater than 100"
fi
```

(ii)
```
#!/bin/bash
#Linux Shell Scripting (if-then-else-fi Shell Script)
echo -n "Input value to check:"
read n
if [ $n -ge 1 -a $n -le 100]
then echo "Number is between 1 & 100."
else echo "Number greater than 100..."
fi
```

output

Enter First number: 34

Enter second number:

56

Enter third number

67

c is greatest..

Expt. No. ______ Date ______ Page No. 16

(iii)
```
#!/bin/bash
#Linux Shell ~~progress~~ Scripting (Logical operators
        with nested if-else constrain)
echo "Enter First number:"
read a
echo "Enter Second number:"
read b
echo "Enter third number:"
read c
if [ $a -gt $b]
then if [$a -gt $b]
then echo 'A is greatest.."
else echo 'E is greatest.."
fi
else if [ $b -gt $c]
then echo "B is greatest.."
else echo "c is greatest.."
fi
fi
```

(e) Write a shell programing to demonstrate logical operators, Logical operators used in shell scripting are-

-gt -> greater than
-lt -> Less than
-ge -> greater than or equal to
-le -> less than or equal to
-ne -> not equals to

Teacher's Signature ______

Output
Enter the chapter name;
VGA
You have a video graphics
adapter..

-eq -> equals to write a shell scripting program to demonstrate else + if equals elif, case structure.

Program.

```
#! /bin/bash
#Linux shell scripting (else-if equals elif)
echo "Enter the adapter name;"
read adapter
if ["$adapter" == MA]
then echo "You have a monochrome
adapter.." elif ["$adapter" == CGA]
then echo "You have a video graphics
adapter.." else echo "You have entered wrong
choice.."
fi
```

(f) Write a shell script to demonstrate the use of case structure

```
#! /bin/Shell
# Linux Shell Scripting (case structure)
echo "Enter weekday in number:"
read week
case $week in
1) echo "You entered for monday'." ;;
2) echo "You entered for tuesday.." ;;
3) echo "You entered for wednesday.."
4) echo "You entered for Thursday.." ;;
```

output

Enter weekday in number.

2

You entered for Tuesday.

Date

Expt. No.


```
5) echo "You entered for Friday.." ;;
6) echo "You entered for Saturday.." ;;
7) echo "You entered for Sunday.." ;;
*) echo "Only 7 days in week.." ;;
esac
```

Viva Question

Q1 What is an interactive shell script?

Ans A script that takes input from the user during execution using command like read.

Q2 ~~Positional Parame'~~

Q2 What are positional parameters in shell programming

Ans Variables like $1, $2, $3 used to access command-line arguments passed to the script.

Q3 Which command line is used for arithematic operations in shell script?

Ans The expr command is commonaly used to perform arithmetic operations in shell script.

Q4 What is the difference between if-then & if-then-else

Ans If-then : executes codes only if condition is true.

If-then-else : executes codes block if true, another if false.

Q5 What is use of case statement in shell programing

Ans Used to perform different actions based on multiple possible values (like switch-case in C).

Teacher's Signature

output

```
Content of output from 5th line
Difference between file 1 & file 2
1c1
< hiiiiiiiiiiiiii helloooooo howrrrr
                  youuuu!!!!---
> how
for matching certain word
enter word which you want to match in
                  file
```


Aim:- Write a shell script to create a file in $USER/class/batch directory. Follow the instructions

(i) Input a page profile to yourself, copy it into other existing file.

(ii) Start printing file at certain line

(iii) Print all the difference two file, copy the two files at $USER/CSC/2007 directory.

(iv) Print lines matching certain word pattern.

Solution:-

```
echo "Content of Profile file"
cat profile #here profile is a file
echo "Content of after existing file named
      output"
cat output
echo "Content after adding profile file into
      output file"
cat profile >> output
cat output

echo "Content of output from 5th line"
tail -n +5 output #or tail --lines=+5 output

echo "Difference between file 1 & file 2"
diff file1 file2

echo "for matching certain word"
```

echo "enter word which you want to match in file"
read word
grep $word output.

Viva Question

Q1 What is the use of pipe (|) in Linux?
Ans A pipe transfers the output of one command as input to another command.

Q2 What does > operator do?
Ans The > operator redirect command output into a file & overwrites existing content.

Q3 What is the function of the kill command?
Ans. The kill command terminates a running process using its Process ID (PID)

Q4 What command displays inode number?
Ans The ls -i command displays inode numbers of files and directories.

Q5 What is the use of the grep command?
Ans The grep command searches & displays lines matching a specific pattern in a file.